



Innovators

“INDIA CAN LEAD IN IP—If It Backs Its Own Builders”

Axiro Semiconductor designs RF and satellite chips, ships millions worldwide, and retains its startup mindset while tackling complex technical challenges. In a conversation with Nidhi Agarwal from EFY, Dr Naveen Yanduru shared insights into the company's global operations, RF front-end innovations, and what India needs to build and scale its own chip products.

DR NAVEEN YANDURU
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What gaps in satellite infrastructure do you think are still not fully addressed?

When it comes to satellite communication, there are three main challenges. First is the G/T, or signal-to-noise performance. Since satellites are far from Earth, systems require very low noise levels to maintain reliable connections, especially given the high path loss. Second is power consumption. Satellite terminals often use large arrays, and if power is not managed effectively, the panels can overheat, and performance can drop. This is especially critical in remote areas with limited access to power. Third is cost. Phased-array systems use many chips, so keeping the overall terminal affordable is key to making services such as LEO constellations commercially viable. These three factors—G/T, power consumption, and cost—are central to the design of competitive satellite communication chips.

How do you handle Doppler shift and fast-moving satellite links at the silicon level?

Doppler shift is a critical challenge, especially as video data traffic increases. In traditional systems, analogue beamformers with phase shifters using true-time delays were used, which can address some Doppler-related issues. Implementing true-time-delay-based phase shifting at the silicon level helps, but it increases terminal cost and size because these components are bulkier and more difficult to design. At the system level, customers also apply techniques to mitigate Doppler effects when communicating with fast-moving satellites.

What are the key technical challenges in building up-down converter chips for satellite links?

Up-down converters are critical because they handle frequency translation, but this comes with several technical challenges. One major issue is managing spurious signals and out-of-band noise to avoid interference in the transmitter. Another challenge is keeping the intermediate frequency low enough for downstream chips to process it efficiently. However, this brings

the image frequency closer, making filtering more difficult. Overall, maintaining low spurious levels, good phase-noise performance, and strong image rejection are the key challenges in designing these chips.

What do you think are the biggest technical bottlenecks in the RF industry?

The RF market is broad; anytime wireless communication is involved, RF technology is required. There are opportunities across satellite communications, cellular networks, wireless infrastructure, handsets, and other applications. The main challenge right now is on the cellular side. While 5G deployments are ongoing, 6G still feels like a distant goal, and the industry has yet to develop a compelling business case for it. That makes it difficult to move to the next wave before the current one matures. However, RF spans so many applications that companies can focus on other markets, such as defence and aerospace, to balance short-term slowdowns in the cellular sector.